A PROJECT TITLE

ON

Farmland Zone Clustering Using K-Means and Hierarchical Clustering

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



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1. Abstract:

Agriculture continues to play a central role in regional economic planning, and understanding spatial variation among farmland parcels is essential for efficient resource allocation and sustainable development. Traditional methods of classifying agricultural land rely heavily on manual assessment, which is time-consuming, subjective, and difficult to scale across large regions. This project presents a data-driven approach to farmland zone clustering using K-Means and Hierarchical Clustering to automatically identify meaningful groupings among farmland parcels. Agricultural data comprising attributes such as soil quality, irrigation availability, crop yield, rainfall, and land area are collected and pre-processed through cleaning, handling of missing values, and normalization so that all features contribute equally to the clustering process. K-Means clustering is applied to partition farmland zones into distinct groups based on similarity in these attributes, offering a fast and computationally efficient way to segment large datasets. In parallel, Agglomerative Hierarchical Clustering is used to validate these groupings and further explore the nested structure of the zones through dendrogram analysis, revealing relationships between clusters at different levels of granularity. The optimal number of clusters for K-Means is determined using the Elbow Method and Silhouette Score, ensuring that the resulting groupings are meaningful and well-separated. The resulting clusters help identify farmland zones with similar agricultural potential, enabling targeted interventions such as customized irrigation planning, fertilizer distribution, crop recommendation strategies, and infrastructure development. A comparative analysis between the two clustering methods evaluates their respective strengths and limitations on farmland datasets of varying size and complexity, providing insight into which technique suits specific agricultural planning scenarios. Visualizations such as cluster scatter plots and dendrograms are used to interpret and communicate the results to non-technical stakeholders. The proposed system ultimately aims to support policymakers, agronomists, and farmers in making informed, data-backed decisions that improve productivity, optimize resource utilization, and promote sustainable land use practices across diverse farmland regions.

2. INTRODUCTION:

Agriculture continues to be the backbone of many regional economies, and the efficient management of farmland resources is critical for maximizing productivity and sustainability. With growing access to agricultural datasets covering soil characteristics, irrigation patterns, rainfall, and crop yield, there is a significant opportunity to apply machine learning techniques to uncover meaningful patterns within farmland zones. Manually classifying farmland based on multiple interacting factors is time-consuming and prone to human bias, which motivates the use of unsupervised learning methods to automatically group similar zones together. This project focuses on applying K-Means clustering and Agglomerative Hierarchical Clustering to segment farmland into meaningful zones based on shared agricultural characteristics. K-Means offers a fast and scalable partitioning approach, while Hierarchical Clustering provides an interpretable, tree-based view of how zones relate to one another at different levels of granularity. By comparing the outcomes of both techniques, this project aims to identify the most effective clustering strategy for farmland zoning and to demonstrate how data-driven insights can support smarter agricultural planning, resource distribution, and policy decisions.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10 / 11 or Linux (Ubuntu 20.04 and above)
- Programming Language: Python 3.8 or above
- IDE / Notebook: Jupyter Notebook, Google Colab, or VS Code
- Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, SciPy
- Clustering Tools: Scikit-learn (KMeans, AgglomerativeClustering, dendrogram)
- Visualization Tools: Matplotlib, Seaborn, Plotly (for cluster and dendrogram plots)
- Data Format: CSV / Excel files containing farmland attribute data

Signature of the Guide **Course Instructor**